Number theory – 5

Equations in integers

Homework

14 сентября 2026 года

Theoretical minimum. $ax + by = c$ is solvable exactly when $\gcd(a, b) \mid c$, and the solutions form an arithmetic progression. Pell's equation $x^2 - dy^2 = 1$ with d a non-square has infinitely many solutions, all powers of the fundamental one, which is read off from the continued fraction of $\sqrt{d}$; the existence follows from Dirichlet's approximation theorem plus pigeonhole. Infinite descent rules out $x^4 + y^4 = z^2$ and hence Fermat's equation for exponent 4. Every integer is a sum of four squares; three squares suffice except for $4^a(8b + 7)$; two squares exactly when every prime $\equiv 3 \pmod 4$ occurs to an even power. Solvability in $\mathbb{Z}$ implies solvability modulo every n, and that implication run backwards is how impossibility is proved; the converse fails in general, the standard counterexample being Selmer's cubic $3x^3 + 4y^3 + 5z^3 = 0$. Thue's theorem: an irreducible homogeneous form of degree at least three takes each value finitely often.

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1. Prove that the equation $x^2 + y^2 + z^2 = 2xyz$ has no solution in positive integers, and solve the Markov equation $x^2 + y^2 + z^2 = 3xyz$ completely by Vieta jumping.

2. Find all pairs (m, n) of non-negative integers for which $m^2 + 2 \cdot 3^n = m(2^{n+1} - 1)$. (IMO shortlist 2010, N2)

3. Find all positive integers n for which there exist non-negative integers $a_1, \ldots, a_n$ with

$$\frac{1}{2^{a_1}} + \frac{1}{2^{a_2}} + \cdots + \frac{1}{2^{a_n}} = \frac{1}{3^{a_1}} + \frac{2}{3^{a_2}} + \cdots + \frac{n}{3^{a_n}} = 1.$$

(IMO shortlist 2012, N7)

4. Prove that for any pair of positive integers k and n there exist k positive integers $m_1, \ldots, m_k$ such that $1 + \frac{2^k - 1}{n} = \left(1 + \frac{1}{m_1}\right) \cdots \left(1 + \frac{1}{m_k}\right)$. (IMO shortlist 2013, N2)

5. Let $n \geqslant 2$ and $A_n = \{2^n - 2^k : 0 \leqslant k < n\}$. Determine the largest positive integer that cannot be written as a sum of one or more, not necessarily distinct, elements of A_n. (IMO shortlist 2014, N1)

6. Solve $x^2 - 2y^2 = \pm 1$ completely and use the solutions to find all triangular numbers that are perfect squares; then show that the equation $x^2 - dy^2 = -1$ is solvable if and only if the continued fraction of $\sqrt{d}$ has odd period.

7. Prove that the equation $y^2 = x^3 + 7$ has no solution in integers, by adding 1 to both sides and factoring modulo 4; then prove that $y^2 = x^3 - 2$ has only $x = 3$, $y = \pm 5$, using unique factorisation in $\mathbb{Z}[\sqrt{-2}]$.

8. Call an integer a *friendly* if the equation $(m^2 + n)(n^2 + m) = a(m - n)^3$ has a solution in positive integers. Prove that there are at least 500 friendly integers in $\{1, \ldots, 2012\}$, and decide whether $a = 2$ is friendly. (IMO shortlist 2012, N4)

9. Prove that the number of representations of n as an ordered sum of four squares equals $8 \sum_{d \mid n,\ 4 \nmid d} d$ (Jacobi), at least in the case of odd n, by comparing the coefficients of $\theta(q)^4$; or, if the generating-function computation is too long, verify the formula for all $n \leqslant 20$ and prove the four-square theorem by descent.

10. Prove that there are infinitely many positive integers that are not the sum of three cubes, and determine the obstruction modulo 9; then explain why the same argument says nothing about the representability of 33 and 42.